Python Core Data Types – MNC Interview Coding Questions

40 High-Frequency MNC Interview Questions (List, Tuple, Set & Dictionary)

Total Questions: 40 **Focus:** Search, Sort, Duplicates, Frequency & Data Logic

🟢 LIST (Questions 1 – 10)

1. Linear Search

Given a list of employee IDs and a target ID, search for the target and display its position.

```
Input:  List = [101, 205, 304, 405, 502], Target = 304
Output: Element found at index 2
```

2. Find Second Largest

Find the second-largest number in a list without using sort().

```
Input:  [45, 12, 89, 67, 34, 90]
Output: 89
```

3. Find Second Smallest

Find the second-smallest element in a list without sorting the complete list.

```
Input:  [45, 12, 89, 67, 34, 90]
Output: 34
```

4. Sort Without Built-in Function

Sort a list of numbers in ascending order without using sort() or sorted().

```
Input:  [64, 34, 25, 12, 22, 11, 90]
Output: [11, 12, 22, 25, 34, 64, 90]
```

5. Remove Duplicates

Remove duplicate elements while maintaining their original order.

```
Input:  [10, 20, 10, 30, 20, 40]
Output: [10, 20, 30, 40]
```

6. Find Missing Number

Given numbers from 1 to N with one number missing, find the missing number.

```
Input:  [1, 2, 3, 5, 6]
Output: 4
```

7. Find Common Elements

Find the common elements between two lists without using set conversion.

```
Input:  A = [10, 20, 30, 40], B = [30, 40, 50, 60]
Output: [30, 40]
```

8. Move Zeros

Move all zeros to the end while maintaining the order of other elements.

```
Input:  [0, 5, 0, 3, 8, 0, 2]
Output: [5, 3, 8, 2, 0, 0, 0]
```

9. Frequency of Elements

Find how many times each element occurs in a list.

```
Input:  [10, 20, 10, 30, 20, 10]
Output: 10 : 3
        20 : 2
        30 : 1
```

10. Stock Price Analysis

A company stores daily stock prices in a list. Find the highest price, lowest price, and average price.

```
Input:  [120.5, 132.0, 118.4, 145.2, 128.9]
Output: Highest: 145.2, Lowest: 118.4, Average: 129.0
```

🟡 TUPLE (Questions 11 – 20)

11. Tuple Search

Given a tuple of employee IDs, search for a particular ID and return its index.

```
Input:  Tuple = (101, 202, 303, 404), Target = 303
Output: Index: 2
```

12. Tuple Sorting

Given a tuple of numbers, create a sorted tuple without modifying the original tuple.

```
Input:  (50, 10, 40, 20, 30)
Output: (10, 20, 30, 40, 50)
```

13. Find Maximum and Minimum

Find the maximum and minimum values from a tuple without using max() and min().

```
Input:  (15, 82, 24, 95, 33)
Output: Max: 95, Min: 15
```

14. Count Duplicate Values

Find the number of times each value occurs in a tuple.

```
Input:  (10, 20, 10, 30, 20, 10)
Output: 10 : 3
        20 : 2
        30 : 1
```

15. Remove Duplicate Values

Create a new tuple containing only unique elements.

```
Input:  (1, 2, 2, 3, 4, 4, 5)
Output: (1, 2, 3, 4, 5)
```

16. Tuple Comparison

Given two tuples, check whether they contain the same elements regardless of order.

```
Input:  T1 = (1, 2, 3), T2 = (3, 2, 1)
Output: Equal Elements: True
```

17. Find Common Elements

Find the common elements between two tuples.

```
Input:  A = (10, 20, 30, 40), B = (30, 40, 50, 60)
Output: (30, 40)
```

18. Student Ranking

Given a tuple containing student marks, find the highest, second-highest, and third-highest marks.

```
Input:  (78, 92, 85, 88, 95, 60)
Output: Top 3: 95, 92, 88
```

19. Tuple of Employee Records

Given employee records as tuples, find the employee with the highest salary.

```
Input:  employees = (("E101", "Arun", 45000), ("E102", "Priya", 52000), ("E103", "Kumar",
48000))
Output: Highest Salary: ('E102', 'Priya', 52000)
```

20. Sort Employee Records

Given employee records as tuples, sort the employees based on salary.

```
Input:  employees = (("E101", "Arun", 45000), ("E102", "Priya", 52000), ("E103", "Kumar",
48000))
Output: (('E101', 'Arun', 45000), ('E103', 'Kumar', 48000), ('E102', 'Priya', 52000))
```

🔵 SET (Questions 21 – 30)

21. Find Common Skills

Two developers have different skill sets. Find their common skills.

```
Input:  A = {"Python", "Java", "SQL", "Git"}, B = {"Python", "SQL", "HTML", "Git"}
Output: {'Python', 'SQL', 'Git'}
```

22. Unique Elements

Given a list containing duplicate values, use a set to find the number of unique elements.

```
Input:  [1, 2, 2, 3, 4, 4, 4, 5]
Output: Unique count: 5
```

23. Find Missing Employee IDs

Given all employee IDs and the IDs that reported attendance, find employees who are absent.

```
Input:  All IDs = {101, 102, 103, 104, 105}, Present = {101, 103, 105}
Output: Absent: {102, 104}
```

24. Find Duplicate Values

Given a list, use set operations to identify which values are duplicated.

```
Input:  [10, 20, 30, 10, 20, 40]
Output: Duplicates: {10, 20}
```

25. Union of Skills

Find all unique skills possessed by two employees combined.

```
Input:  A = {"Python", "SQL"}, B = {"Java", "SQL", "Git"}
Output: {'Python', 'SQL', 'Java', 'Git'}
```

26. Set Difference

Find skills that Employee A has but Employee B does not have.

```
Input:  A = {"Python", "Java", "SQL"}, B = {"SQL", "Git"}
Output: {'Python', 'Java'}
```

27. Course Registration

Two groups registered for Python and Java. Find both, Python-only, Java-only, and total unique students.

```
Input:  Python = {"Arun", "Priya", "Kumar"}, Java = {"Priya", "Divya", "Rahul"}
Output: Both: {'Priya'}
        Python-Only: {'Arun', 'Kumar'}
        Java-Only: {'Divya', 'Rahul'}
        Total Unique: 5
```

28. Unique Visitors

A website records visitor IDs multiple times during a day. Find the total number of unique visitors.

```
Input:  [101, 102, 101, 103, 104, 102, 105]
Output: 5
```

29. Subset Validation

Check whether all required skills for a job are available in a candidate's skill set.

```
Input:  Required = {"Python", "SQL", "Git"}, Candidate = {"Python", "SQL", "Git", "Java"}
Output: Eligible
```

30. Common Products

Two stores maintain sets of product IDs. Find the products available in both stores.

```
Input:  Store1 = {"P101", "P102", "P103"}, Store2 = {"P102", "P103", "P104"}
Output: {'P102', 'P103'}
```

🔴 DICTIONARY (Questions 31 – 40)

31. Employee Search

Given an employee dictionary, search for an employee ID and display the employee name.

```
Input:  employees = {101: "Arun", 102: "Priya", 103: "Kumar"}, Target = 102
Output: Priya
```

32. Find Highest Salary

Given employee names and salaries in a dictionary, find the employee with the highest salary.

```
Input:  {"Arun": 45000, "Priya": 52000, "Kumar": 48000, "Divya": 60000}
Output: Divya with salary 60000
```

33. Sort Dictionary by Values

Sort employees according to their salary from lowest to highest.

```
Input:  {"Arun": 45000, "Priya": 52000, "Kumar": 48000, "Divya": 60000}
Output: [('Arun', 45000), ('Kumar', 48000), ('Priya', 52000), ('Divya', 60000)]
```

34. Sort Dictionary by Keys

Sort a dictionary of employee names and IDs alphabetically by employee name.

```
Input:  {"Priya": 102, "Arun": 101, "Kumar": 103}
Output: {'Arun': 101, 'Kumar': 103, 'Priya': 102}
```

35. Word Frequency

Given a sentence, count the frequency of every word using a dictionary.

```
Input:  "python java python sql java python"
Output: python : 3
        java : 2
        sql : 1
```

36. Character Frequency

Given a string, count the frequency of each character using a dictionary.

```
Input:  "programming"
Output: p:1, r:2, o:1, g:2, a:1, m:2, i:1, n:1
```

37. Highest Frequency Element

Given a list, use a dictionary to find the element that occurs most frequently.

```
Input:  [10, 20, 10, 30, 20, 10]
Output: 10
```

38. Group Employees by Department

Given employee department assignments, group employees according to their department.

```
Input:  {"Arun": "IT", "Priya": "HR", "Kumar": "IT", "Divya": "Finance", "Rahul": "IT"}
Output: {'IT': ['Arun', 'Kumar', 'Rahul'], 'HR': ['Priya'], 'Finance': ['Divya']}
```

39. Product Inventory

Given product stock quantities, find highest stock, lowest stock, total stock, and items below 10 stock.

```
Input:  {"Laptop": 15, "Mouse": 8, "Keyboard": 25, "Monitor": 5}
Output: Highest: Keyboard, Lowest: Monitor, Total: 53, Below 10: ['Mouse', 'Monitor']
```

40. Student Ranking ⭐

Find highest/lowest scorer, sort students by marks, and display the top 3 students.

```
Input:  {"Arun": 85, "Priya": 92, "Kumar": 78, "Divya": 95, "Rahul": 88}
Output: Highest: Divya (95), Lowest: Kumar (78)
        Sorted: [('Divya', 95), ('Priya', 92), ('Rahul', 88), ('Arun', 85), ('Kumar', 78)]
        Top 3: [('Divya', 95), ('Priya', 92), ('Rahul', 88)]
```